SECTION 1 IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY

1.1 Product identifiers

Product Form	Substance
Trade Name	Lithium Nickel Manganese Cobalt Oxide (LiNiMnCoC) Powder (NMC 111)
Product Number	NG08BE14067
CAS Number	346417-97-8

1.2 Relevant identified uses of the substance or mixture and uses adviced against

Use of the substance/mixture	Research

1.3 Details of the supplier of safety data sheet

Company	Nanografi Nanotechnology
Address	ODTÜ Teknokent İkizler Binası B-1/H ODTÜ Teknokent 06531- ANKARA
Phone	+90 312 285 85 09
Fax	+90 312 210 13 09

1.4 Emergency Telephone Number

Emergency Telephone Number	+90 312 285 85 09

SECTION 2 HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008

H317	Skin Sensitizing (1A)
H351	Carcinogenicity (2)

For the full text of the H-Statements mentioned in this Section, see Section 16.

2.2 Label Elements

Hazard Pictograms

Signal Word	Warning
Hazard Statements	H317 May cause an allergic skin reaction. H351 Suspected of causing cancer.
Precautionary Statements	P201 Obtain special instructions before use. P202 Do not handle until all safety precautions have been read and understood. P261 Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray. P272 Contaminated work clothing should not be allowed out of the workplace. P280 Wear protective clothes. P302 + P352 IF ON SKIN: Wash with plenty of soap and water. P308 + P313 IF exposed or concerned: Get medical advice/attention. P333 + P313 If skin irritation or rash occurs: Get medical advice/attention. P363 Wash contaminated clothing before reuse. P501 Dispose of contents/ container to an approved waste disposal plant.

2.3 Other Hazards

None.

SECTION 3 COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substances

Synonyms	Lithium Nickel Manganese Cobalt Oxide, Cobalt Lithium Manganese Nickel Oxide
CAS Number	346417-97-8

3.2 Mixtures

Not applicable.

SECTION 4 FIRST AID MEASURES

4.1 Description of first aid measures

First-aid measures after inhalation	Remove victim to fresh air. If required provide artificial respiration. Consult a physician.
First-aid measures after skin contact	Wash with plenty of soap and water. Consult a physician.
First-aid measures after eye contact	Flush eyes with water as a precaution.
First-aid measures after ingestion	Rinse mouth with water. Consult a physician.

4.2 Most important symptoms and effects, including acute and delayed

The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11.

4.3 Indication of any immediate medical attention and special treatment needed

No data available.

SECTION 5 FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable Extinguishing Media	Dry Chemical Powder, alcohol-resistant foam, water spray, carbon dioxide.
Unsuitable Extinguishing Media	No data available.

5.2 Special hazards arising from the substance or mixture

Hazardous Decomposition Products	Nickel, Nickel oxides, Lithium oxides, Cobalt, Cobalt oxides, Manganese, Manganese oxides

5.3 Advice for firefighters

In case of fire wear self-contained breathing apparatus.

SECTION 6 ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Protective Equipment	Wear suitable protective clothing.
Emergency Procedures	Avoid formation of dust. Avoid breathing vapors, mist or gas. Ensure adequate ventilation. Evacuate area.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not allow product to enter drains.

6.3 Methods and material for containment and cleaning up

Pick up and arrange disposal without creating dust. Sweep up and shovel. Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

For further information refer to section 13.

SECTION 7 HANDLING AND STORAGE

7.1 Precautions for safe handling

Precautions for Safe Handling	Avoid contact with skin, eyes and clothes. Avoid formation of dust and aerosols. Avoid formation of dust and aerosols. Provide appropriate exhaust ventilation at places where dust is formed.

7.2 Conditions for safe storage, including any incompatibilities

Storage Conditions	Store in dry, cool and well-ventilated area. Keep container tightly closed.

Technical Measures	None.

7.3 Specific end use(s)

Apart from the uses mentioned in section 1.2 no other specific uses are stipulated.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Ingredient	Authority	Type	Limit
LiNiCoMnO2	ACGIH TLV	TWA	0.02 mg/m3
Remarks	Pulmonary function, Asthma, Myocardial effects Substances for which there is a Biological Exposure Index or Indices (see BEI® section) Confirmed animal carcinogen with unknown relevance to humans varies		

8.2 Exposure controls

Appropriate Engineering Controls	Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.
Hand Protection	Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands. The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it.
Eye Protection	Safety glasses with face shield conforming to EN166.
Skin and Body Protection	Impervious clothing. The type of protective equipment must be selected according to the concentration and

	amount of the dangerous substance at the specific workplace.
Respiratory Protection	Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).
Environmental Exposure Controls	Avoid release to the environment.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Physical State	Solid
Appearance	Powder
Color	Dark Brown - Black
Odor	No data available
Odor Threshold	No data available
pH	No data available
Relative Evaporation Rate	No data available
Melting Point	No data available
Freezing Point	No data available
Boiling Point	No data available
Flash Point	No data available
Auto-ignition Temperature	No data available
Decomposition Temperature	No data available
Flammability (solid, gas)	No data available
Vapor Pressure at 20°C	No data available
Relative vapor density at 20°C	No data available
Relative Density	No data available

Solubility in H2O	No data available
Viscosity, Kinematic	No data available
Viscosity, Dynamic	No data available
Explosive Properties	No data available
Oxidizing Properties	No data available

9.2 Other information

No other information available.

SECTION 10 STABILITY AND REACTIVITY

10.1 Reactivity

No data available.

10.2 Chemical stability

Stable at normal conditions.

10.3 Possibility of hazardous reactions

No data available.

10.4 Conditions to avoid

No data available.

10.5 Incompatible materials

Strong oxidizing agents.

10.6 Hazardous decomposition products

Hazardous decomposition products formed under fire conditions. - Nickel, Nickel oxides, Lithium oxides, Cobalt, Cobalt oxides, Manganese, Manganese oxides

Other decomposition products - No data available

In the event of fire: see section 5.

SECTION 11 TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute Toxicity	No data available
Skin corrosion/irritation	No data available
Serious eye damage/irritation	No data available
Respiratory or Skin Sensitization	Skin Sensitizing Category 1A
Germ Cell Mutagenicity	No data available.
Carcinogenity	Limited evidence of a carcinogenic effect.
Reproductive Toxicity	No data available
STOT-Single Exposure	No data available
STOT-Repeated Exposure	No data available
Aspiration Hazard	No data available
Other Information	RTECS: Not available Large doses of lithium ion have caused dizziness and prostration, and can cause kidney damage if sodium intake is limited. Dehydration, weight loss, dermatological effects, and thyroid disturbances have been reported. Central nervous system effects that include slurred speech, blurred vision, sensory loss, ataxia, and convulsions may occur. Diarrhea, vomiting, and neuromuscular effects such as tremor, clonus, and hyperactive reflexes may occur as a result of repeated exposure to lithium ion. Bone marrow - Irregularities - Based on Human Evidence To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

SECTION 12 ECOLOGICAL INFORMATION

12.1 Toxicity

Ecology-general	No data available
Acute aquatic toxicity	No data available
Chronic aquatic toxicity	No data available

12.2 Persistance and degradability

No data available.

12.3 Bioaccumulative potential

No data available.

12.4 Mobility in soil

No data available.

12.5 Results of PBT and vPvB assessment

PBT/vPvB assessment not available as chemical safety assessment not required/not conducted.

12.6 Other adverse effects

No data available.

SECTION 13 DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Product	Offer surplus and non-recyclable solutions to a licensed disposal company. Material may be mixed with a combustible solvent and burned in a chemical incinerator equipped with an afterburner and scrubber.
Contaminated Packaging	Dispose of as unused product.

SECTION 14 TRANSPORT INFORMATION

Safe to carry by any means of transportation. Not restricted any mode of transportation.

SECTION 15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations

SARA 302 Components

No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.

SARA 311/312 Hazards

Acute Health Hazard. Chronic Health Hazard

SARA 313 Components

This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.

Massachusetts Right to Know Components

No components are subject to the Massachusetts Right to Know Act.

California Prop. 65 Components

This product does not contain any chemicals known to State of California to cause cancer, birth defects, or any other reproductive harm.

Pennsylvania Right to Know Components

Lithium Nickel Manganese Cobalt Oxide, CAS#: 346417-97-8

SECTION 16 OTHER INFORMATION

16.1 Abbreviation of acronyms

CLP	Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation (EC) No 1907/2006

PBT	Persistent Bioaccumulative Toxic
vPvB	Very Persistent and Very Bioaccumulative

16.2 Full text of H-statements

H317	May cause an allergic skin reaction.
H351	Suspected of causing cancer.

Disclaimer

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore beconstrued as guaranteeing any specific property of the product.